Task 3 – Deep-Dive Analysis & Interactive Dashboarding

Customer Segmentation Deep-Dive Analysis

Data Analytics Internship – ApexPlanet Software Pvt. Ltd.

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Internship: Data Analytics | Task 3

1. Introduction

This report presents the deep-dive analysis completed for Task 3 of the Data Analytics Internship. The analysis focuses on customer segmentation using sales and customer-level information. A Power BI dashboard was developed to help explore customer value, sales contribution, order activity, and category-level performance through interactive filters.

2. Business Objective

The main objective is to identify different customer value groups and understand how each group contributes to sales and order activity. The analysis is designed to support business decisions such as identifying high-value customers, understanding customer contribution, and exploring performance across categories, products, cities, genders, and dates.

3. Dataset Overview

Field	Purpose
Order_ID	Unique order identifier
Order_Date	Date of the order
Customer_ID	Customer identifier
Customer_Name	Customer name
Age	Customer age
Gender	Customer gender
Product	Product purchased
Category	Product category
Quantity	Quantity purchased
Unit_Price	Price per unit
Total_Sales	Total sales value
City	Customer/order city

4. Core KPIs

KPI	Formula / Definition	Business Rationale
Total Sales	SUM(Total_Sales)	Measures overall revenue generated.
Total Quantity	SUM(Quantity)	Measures total units/items sold.
Average Order Value	Total Sales ÷ Distinct Orders	Shows the average value generated per order.
Average Unit Price	AVERAGE(Unit_Price)	Shows the average selling price per unit.
Unique Customers	DISTINCTCOUNT(Customer_ID)	Measures the number of distinct customers.

5. Deep-Dive Area: Customer Segmentation

Customer Segmentation was selected as the deep-dive business area because customer-level sales data can be used to distinguish customers according to their contribution. This makes it possible to compare customer groups rather than looking only at overall sales.

6. Segmentation Methodology

A customer-level summary was created containing Customer ID, Customer Name, Total Sales, and Order Count. The average customer sales value was then calculated. Customers were classified using business rules:

- **High Value:** Customer sales ≥ 1.5 × average customer sales.
- **Medium Value:** Customer sales ≥ 0.75 × average customer sales but below the High Value threshold.
- **Low Value:** Customer sales below the Medium Value threshold.

7. Interactive Dashboard

The Power BI solution contains an overall Sales Data Analysis dashboard and a Customer Segmentation & Interactive Analysis dashboard. The deep-dive page provides segment-level analysis and interactive filtering.

Main dashboard capabilities:

- Customer count by customer segment.
- Sales contribution by customer segment.
- Average Order Value by customer segment.
- Order activity by customer segment.
- Category-level analysis with customer segments.
- Interactive filtering by Gender, City, Category, Product, and Date.
- Customer-level sales analysis and Top Customer analysis.

8. Analysis & Key Insights

- Customer segmentation makes it easier to distinguish customers according to their sales contribution.
- The High Value segment represents customers with substantially higher sales contribution according to the defined business rule.
- Comparing sales and order activity by segment helps identify whether a segment contributes through higher-value orders or higher order frequency.
- Category and customer-segment analysis helps identify which product categories are important for different customer groups.

- Interactive filters allow users to investigate the customer segments across different cities, genders, products, categories, and dates.
- Top-customer analysis provides a focused view of customers contributing strongly to sales.

9. Business Recommendations

- Prioritize retention and relationship-building activities for High Value customers.
- Study the purchasing behavior of High Value customers to identify products and categories that drive their contribution.
- Use targeted offers or engagement strategies to encourage Medium Value customers to increase their activity.
- Monitor Low Value customers and identify opportunities to improve engagement without using the same strategy for every customer.
- Use the interactive dashboard regularly to compare customer segments across time, location, product, and category.

10. Tools & Technologies

Power BI: Interactive dashboarding, KPI analysis, customer segmentation, filters, and visualizations.
MySQL / SQL: Business-oriented data analysis and query-based exploration from the earlier task.
Excel: Dataset handling and preparation where applicable.

11. Conclusion

The Task 3 deep-dive analysis successfully extends the earlier exploratory analysis by focusing on customer segmentation. The Power BI dashboard provides an interactive way to compare customer groups, sales contribution, order activity, and category performance. The analysis can help a business understand customer value and make more targeted decisions based on customer behavior.

12. Deliverables

- Power BI interactive dashboard.
- Customer segmentation analysis.
- Core KPI definitions and calculations.
- Deep-dive analysis report.
- LinkedIn demonstration video.
- GitHub repository containing the report and supporting project files.

Note: Dashboard-specific numerical findings should be quoted directly from the final Power BI dashboard when presenting the project, so that the report and live dashboard remain consistent.